Chapter 10 – LU Factorization

$$Ax = b$$

- Gaussian elimination is inefficient when solving systems with same A but different b.
- LU factorization seperates the time consuming elimination of A from the manipulations of the right hand side b.
- Once you factorize A, multiple b case can be handled efficiently.
- Gauss elimination can be expressed as LU factorization

Overview of LU factorization

$$Ax = b$$

$$Ax - b = 0 \quad (1)$$

- Assume (1) could be expressed as an upper triangular system:

$$\underbrace{\begin{bmatrix} u_{11} & u_{12} & u_{13} \\ 0 & u_{22} & u_{23} \\ 0 & 0 & u_{33} \end{bmatrix}}_{U} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \underbrace{\begin{bmatrix} d_1 \\ d_2 \\ d_3 \end{bmatrix}}_{d} \quad (2)$$

$$Ux = d$$

$$Ux - d = 0$$

- Assume that there's a lower diagonal matrix

$$L = \begin{bmatrix} 1 & 0 & 0 \\ l_{21} & 1 & 0 \\ l_{31} & l_{32} & 1 \end{bmatrix} \quad (3)$$

- Premultiply (2) by L, the result is (1)

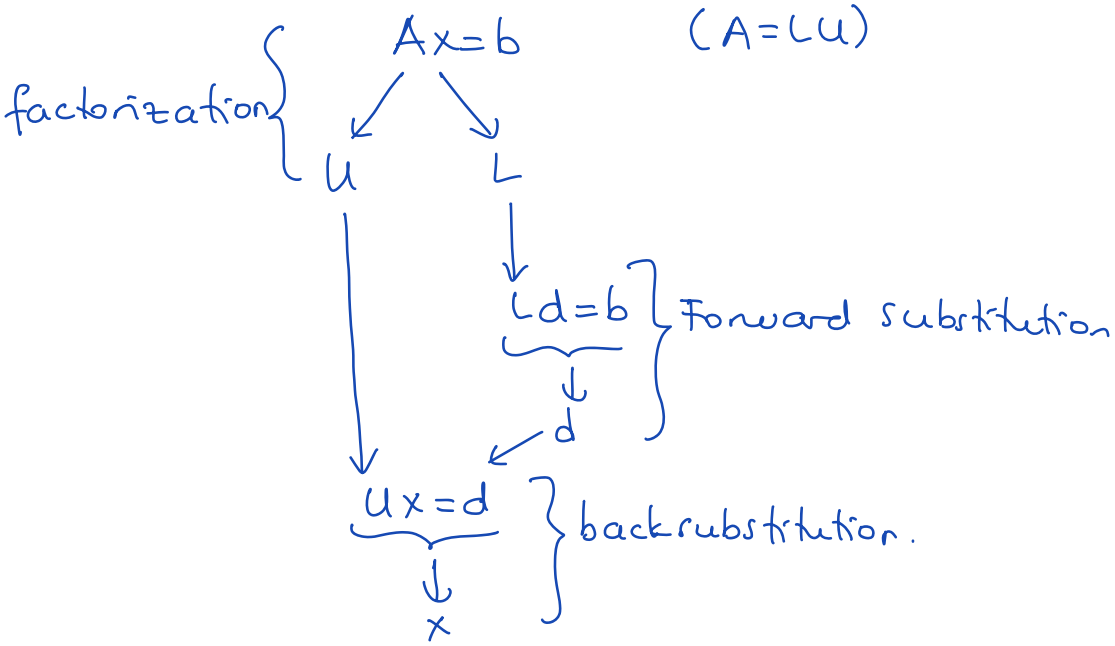
$$L(Ux-d) = Ax-b \quad (4)$$

$$\Rightarrow LU = A \quad (5)$$

$$Ld = b \quad (6)$$

Two-step method:

1) LU factorization of A
2) Substitution step: use L and U to solve for x.
   - Use (6) ($Ld=b$) to solve for intermediate d by forward substitution
   - Use (2) ($Ux=d$) to solve for x using back-substitution.



Gaussian Elimination as LU factorization

- Forward elimination reduces A to an U matrix

$$U = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}$$: desired U format

- L is also produced during the steps:

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

First step: { multiply row 1 by: $f_{21} = \frac{a_{21}}{a_{11}}$
subtract from row 2 to eliminate a_{21}

{ multiply row 1 by: $f_{31} = \frac{a_{31}}{a_{11}}$
subtract from row 3 to eliminate a_{31}

{ multiply row 2 by: $f_{32} = \frac{a'_{32}}{a'_{22}}$
subtract from row 3 to eliminate a'_{32}

- Store the factors in another matrix

$$L = \begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix}$$

$$A = LU$$

$$A = \begin{bmatrix} 1 & 0 & 0 \\ f_{21} & 1 & 0 \\ f_{31} & f_{32} & 1 \end{bmatrix} \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ 0 & a'_{22} & a'_{23} \\ 0 & 0 & a''_{33} \end{bmatrix}$$

$\frac{a_{21}}{a_{11}}$

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & \bullet & \\ & & \end{bmatrix} \quad \bullet : a_{12} \cdot \frac{a_{21}}{a_{11}} + a'_{22} = a_{22}$$

EX Use Naive Gauss elim. to factor the following system into LU decomposition:

$$7x_1 + 2x_2 - 3x_3 = -12$$
$$2x_1 + 5x_2 - 3x_3 = -20$$
$$x_1 - x_2 - 6x_3 = -26$$

$$A = \begin{bmatrix} 7 & 2 & -3 \\ 2 & 5 & -3 \\ 1 & -1 & -6 \end{bmatrix} = \underbrace{\begin{bmatrix} 1 & 0 & 0 \\ 2/7 & 1 & 0 \\ 1/7 & -0.29 & 1 \end{bmatrix}}_{L} \underbrace{\begin{bmatrix} 7 & 2 & -3 \\ 0 & 4.42 & -2.14 \\ 0 & 0 & -6.19 \end{bmatrix}}_{U}$$

$$f_{21} = \frac{2}{7}$$

$$f_{31} = \frac{1}{7}$$

$$f_{32} = -0.29$$

LU Factorization with Pivoting

- In Gaussian elimination partial pivoting is needed.
- Use permutation matrices:

1) Elimination

$$PA = LU$$

↳ permutation matrix

2) Forward elimination

$$PAx = LUx = Pb$$

$$\underbrace{Ux}_{d}$$

$$\begin{pmatrix} Ax = b \\ PAx = Pb \end{pmatrix}$$

Solve $Ld = Pb$ for d

3) Back substitution: Solve $Ux = d$ for x

Ex $\begin{bmatrix} 0.0003 & 3 \\ 1 & 1 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 2.0001 \\ 1.0000 \end{bmatrix}$ Compute LU factorization and find the solution.

a_{11} is close to zero $\Rightarrow$ switch rows:

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \xrightarrow{\text{switch rows}} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$$

$$PA = \begin{bmatrix} 1 & 1 \\ 0.003 & 3 \end{bmatrix} \rightarrow$$ multiply row 1 by 0.003 and subtract from row 2 to eliminate

$$U = \begin{bmatrix} 1 & 1 \\ 0 & 2.9997 \end{bmatrix}, \quad L = \begin{bmatrix} 1 & 0 \\ 0.0003 & 1 \end{bmatrix}$$

$$Pb = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 2.0001 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 2.0001 \end{bmatrix}$$

b

$PAx = Pb$

$LUx = Pb$

$\underbrace{Ux}_{d} = Pb$

$Ld = Pb \Rightarrow$ forward subst.

$$\begin{bmatrix} 1 & 0 \\ 0.0003 & 1 \end{bmatrix} \begin{bmatrix} d_1 \\ d_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 2.0001 \end{bmatrix} \Rightarrow d_1 = 1, \quad d_2 = 1.9998$$

$Ux = d \Rightarrow$

$$\begin{bmatrix} 1 & 1 \\ 0 & 2.9997 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \end{bmatrix} = \begin{bmatrix} 1 \\ 1.9998 \end{bmatrix}$$

back subst.

$$\begin{cases} x_2 = \dfrac{1.9998}{2.9997} = 0.6667 \\ x_1 = \dfrac{1 - 0.6667}{1} = 0.3333 \end{cases}$$

MATLAB Function $[L,U] = lu(X)$

To Solve $Ax = b$

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>> [L,U] = lu(A);
>> d = L\b;
>> x = U\d;
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